FIRST REVIEW

NeuroStamp

Secure Copyright Protection Using DWT-SVD Watermarking

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PROBLEM DEFINITION [1/2]

- **Key Challenges**
- Unauthorized Distribution: Digital images can be copied and shared without owner consent
- Copyright Infringement: Difficulty in proving ownership and tracking violations
- Image Manipulation: Watermarks must survive compression (JPEG), noise, cropping, and other attacks

PROBLEM DEFINITION [2/2]

- Invisibility Requirement: Protection must not degrade visual quality
- Verification Complexity: Need for secure, reliable ownership proof mechanism
- **Solution Approach:**

  Implement a robust, invisible watermarking system that embeds ownership data into the mathematical structure of images, surviving common attacks while maintaining visual fidelity.

OBJECTIVE OF WORK [1/2]

Core Goal

Develop a secure digital watermarking platform that protects intellectual property through invisible, robust watermark embedding.

Technical Goal

Implement DWT-SVD algorithm for watermark embedding and extraction with resilience against common attacks.

OBJECTIVE OF WORK [2/2]

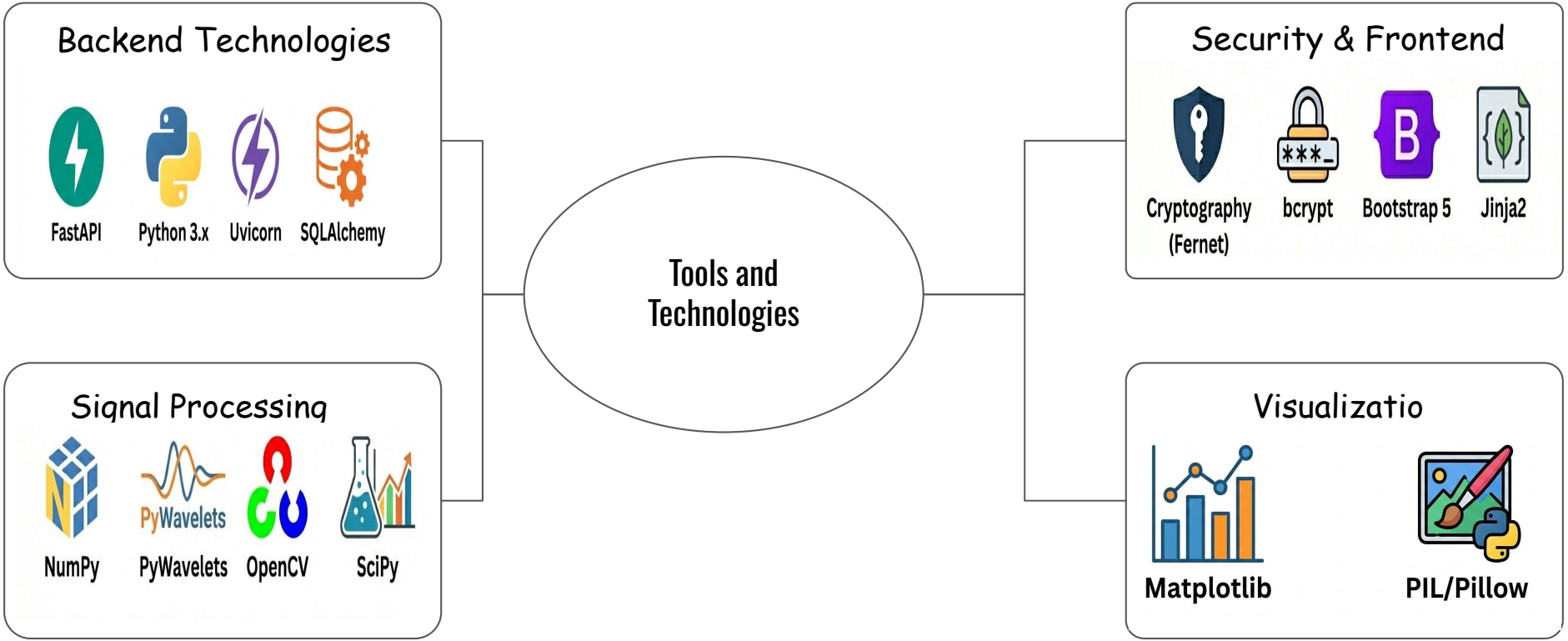
Specific Deliverables

- Block-based DWT-SVD watermarking engine
- Secure user authentication and encrypted watermark key storage
- Web-based interface for stamping, verification, and visualization
- Robustness against JPEG compression (Quality 50+) and Gaussian noise
- Interactive visualization engine to demonstrate algorithm internals
- Perceptual hashing (dHash) for duplicate image detection

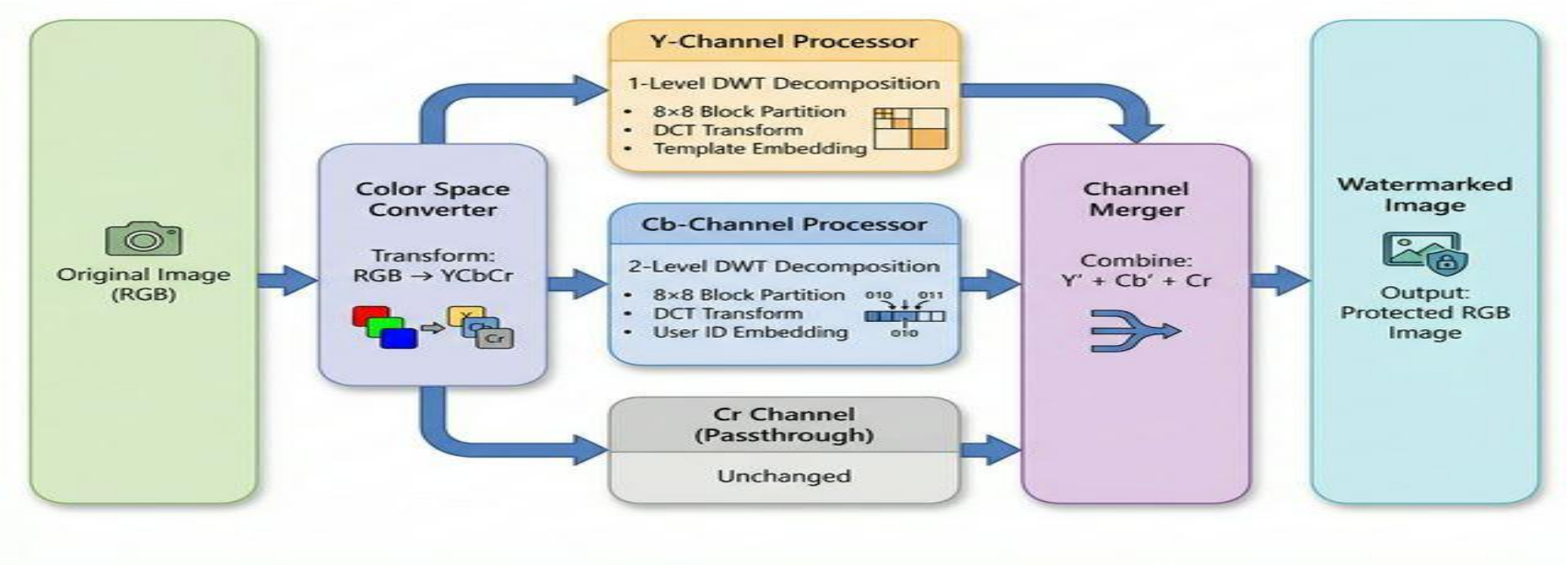
FEASIBILITY ANALYSIS

Aspect	Analysis
Algorithm Complexity	DWT-SVD is well-established in research with proven robustness
Computational Cost	Block-based approach (4×4) reduces processing time vs. full-image SVD
Storage Requirements	Encryption keys stored as serialized arrays (moderate space
Library Support	Python ecosystem has mature libraries (PyWavelets, NumPy, OpenCV)
Web Deployment	FastAPI provides async API with easy deployment options

Tool Specification



METHODOLOGY [1/3]



METHODOLOGY [2/3]

- **Embedding Process**

1. Color Space Conversion : RGB ->YCbCr and extract Y for processing
2. Level-1 DWT Decomposition : LL , LH, HL, HH
3. Block Division : Divide LL subband into non-overlapping 4×4 pixel blocks
4. SVD & Embedding : Compute SVD(block) modify S[0] = S[0] + (α × bit)
5. Reconstruction : Produce watermarked image.

METHODOLOGY [3/3]

- **Watermark Extraction Algorithm**

1. Apply DWT to potentially watermarked image → Extract LL subband
2. Divide into 4×4 blocks and compute SVD for each → Get S'[0]
3. Bit Extraction Rule: diff = S'[0] - S_original[0]

   (If diff > (α/2) → bit = 1, else → bit = 0)
4. Convert binary string back to text watermark

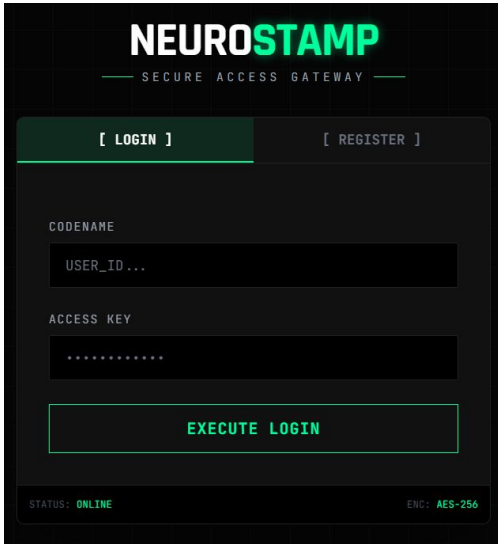
DATA COLLECTION

- Test Data:Images:

1. Standard Images: Lena, Baboon (512×512) for algorithm validation
2. User Uploads: Database stores encrypted watermark keys + metadata
3. Attack Simulations: JPEG compression (Q=50), Gaussian noise (σ=15)

- Progress (60%):
  - Environment & Tools set up.
  - Math Logic (DWT-SVD) working.
  - Web Interface (Upload/Download) active.

PRACTICAL IMPLEMENTATION [1/2]



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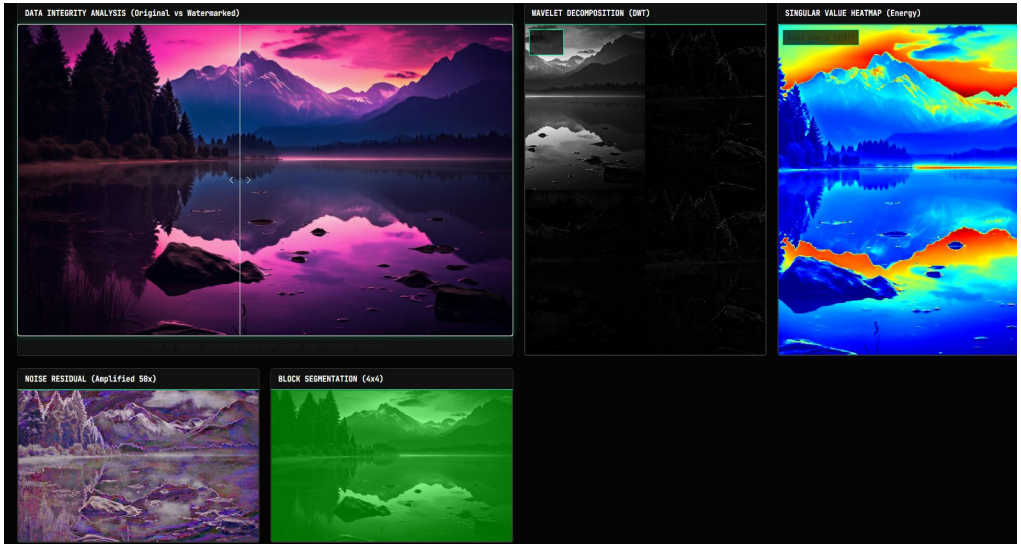
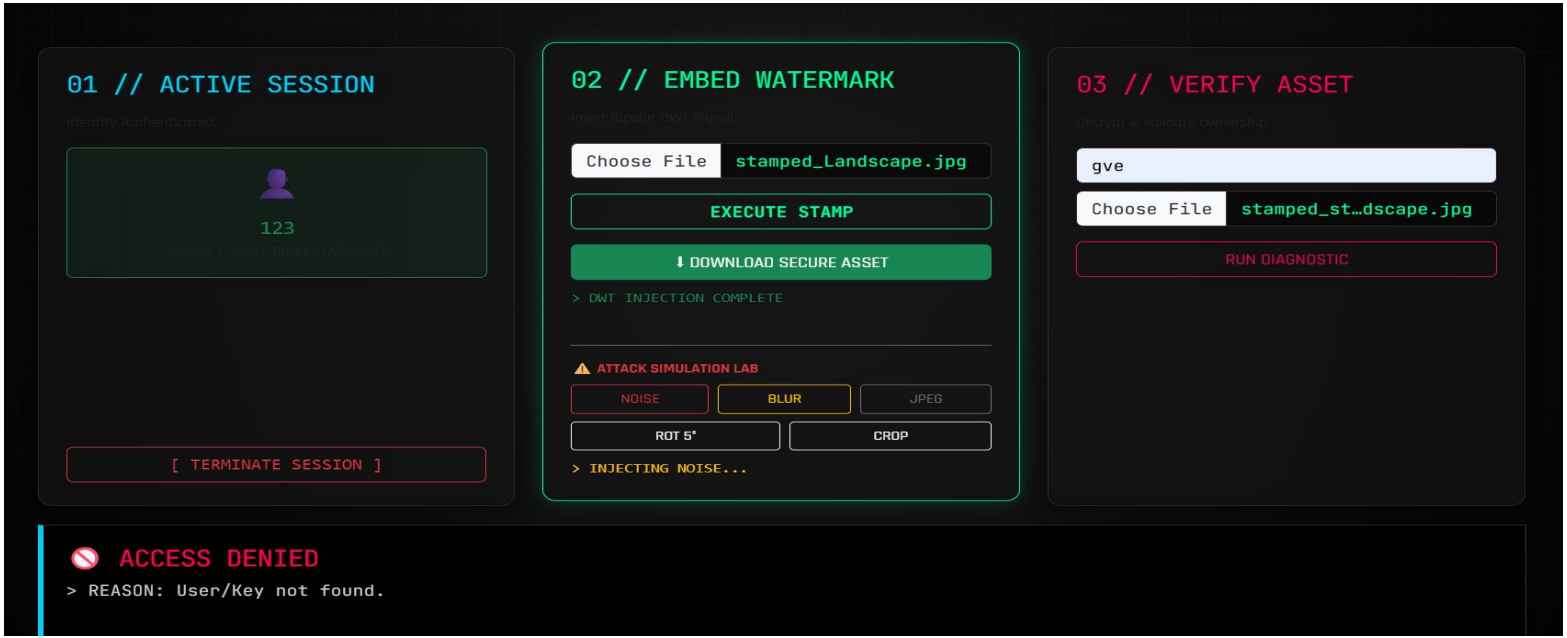


VISUALIZATION ENGINE

PRACTICAL IMPLEMENTATION [2/2]



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REFERENCES

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2 Dong, Y., Yan, R., & Yin, C., 2024, "An adaptive robust watermarking scheme based on chaotic mapping," Scientific Reports, vol. 14, no. 1, pp. 24735, Nature Portfolio.